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CONTROLLING WEBSITES THROUGH HAND GESTURES USING TENSORFLOW JS AND BROWSER EXTENSION

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ABSTRACT

Gesture-based real-time recognition systems have been widely studied for efficient human-computer interaction. However, existing applications that use hand gestures to control websites suffer from several issues, including privacy concerns, high bandwidth requirements, system dependency, and limited configurability. In this paper, we present a novel approach to develop a hand gesture recognition system within the browser, utilising JavaScript deep learning models. Our proposed system uses MediaPipe Hands, a neural network model developed using TensorFlow.js, and a mathematical model to accurately recognize hand gestures in real-time. Our system is easily accessible and portable through a Chrome extension, allowing users to configure custom gestures for website actions. Our proposed approach can enable secure and efficient interaction with websites using hand gestures, overcoming the limitations of existing systems.

Keywords: Hand Gestures, Website Control, Browser Extension, Machine Learning, Tensorflow JS.

I. INTRODUCTION

The past few years have seen an increase in interest in developing gesture-based real-time recognition systems, which have the potential to significantly improve human-computer interaction in various environments. Websites have become the most widely used communication, information sharing, and business platform due to the widespread use of the internet. However, existing hand gesture control applications for websites face various limitations, such as privacy concerns, high bandwidth requirements, system dependency, and lack of configurability.

In response to these challenges, we propose a solution that uses deep learning with JavaScript and browser extensions to develop a hand gesture recognition system within the browser. This solution offers several advantages, such as the ability to learn and recognize complex patterns in data, leading to improved gesture recognition accuracy. Additionally, by implementing the system within a browser extension, we increase its accessibility to users while also providing greater configurability. Furthermore, keeping the system within the browser reduces the potential for system dependency issues.

Our primary objective is to overcome the limitations of existing hand gesture control applications for websites by providing a more seamless and better user experience. Our solution leverages deep learning with JavaScript and browser extensions, which allow the system to learn and adapt over time while prioritizing user privacy and security.

II. RELATED WORK

The previous implementation of gesture control technology was primarily focused on catering to the needs of specific software and applications, resulting in a system-oriented approach. The gesture control solutions were designed to support only limited gesture usage, and were thus less flexible in terms of their application. As a result, the usability of the gesture control technology was significantly reduced, making it less user-friendly. Moreover, these limited gesture control solutions were not able to cater to the diverse needs of users, which limited their appeal and potential usage. This lack of flexibility made it difficult to introduce new gestures or modify the existing ones to meet the requirements of different applications. The challenges associated with the previous implementation of gesture control technology called for a new approach that would be more user-friendly, flexible, and capable of supporting a wider range of gestures to enhance usability

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1. Media Control Using Hand Gestures

In this paper, a software solution for controlling media through hand gestures is presented. The software is written in Python 3.8 and utilizes libraries such as OpenCV and PyAutoGUI to receive input from the computer's webcam and recognize gestures. The user interface is built using the PyQt5 library. The software allows for control of PowerPoint presentations, Portable Document Format (PDF) files, and other media without the need for a mouse or other pointing device. The software is lightweight and easy to build, without the requirement for any external hardware. The software splits the screen into nine parts and detects the object's position on the screen to perform actions. While the system is dependent, non-customizable, and has limited gestures, it provides an advantage of allowing control of media without physical contact with the computer, making it easier to move around while giving presentations or reading documents. The technologies used include OpenCV, PyAutoGUI, NumPy, and PyQt5.

2. Controlling Media Player Using Hand Gesture With VLC Media Player

This paper aims to explore the use of hand gestures to control media players, specifically the VLC media player, with the help of OpenCV and Python. The paper highlights the need for unrestricted interaction between humans and computers, which has made traditional input devices challenging to use. Hand gestures are an important component of body language, and using them as a device for human-computer interaction makes the process more intuitive and interesting. Gesture recognition has gained importance in various applications such as gaming, robot control, and media player control. The use of hand gestures makes interaction easy, and convenient and does not require any extra devices. The paper concludes by noting that while vision and audio recognition can be used together, audio commands may not work well in noisy environments.

3. Controlling Devices Using Hand Gestures

The primary objective of this research is to introduce a novel hand gesture module system that eliminates the need for manual monitoring of switches and indicators. Gestures are actions made with body parts, such as hands or faces, to emphasize speech. These actions can be categorized as either dynamic or static gestures. Gesture recognition is the process of interpreting human actions using computing devices. The aim of this system is to enable the control of electronic devices through hand movements, acting as a remote control for everyday electronic devices such as tube lights and fans. Instead of manual control, this system utilizes hand gestures to operate home appliances. Common electronic equipment found in households, such as TVs, CD players, air conditioners, DVD players, and music systems, can be operated using this gesture-based system. To achieve this, ultrasonic sensors are placed on top of a monitor, and their readings of the distance between the monitor and hand gestures are processed using an Arduino. The Arduino can be connected to a PC/laptop for power and communication. Leap Motion controllers are also employed, a new interactive device that primarily detects hand gestures and finger positions. The study evaluates the performance of the Leap Motion Controller in comparison to a traditional mouse device, using Fitts' law in the evaluation system.

4. Controlling Computer Using Hand Gestures

The paper describes an approach to recognize hand gestures for performing different activities using a deep learning model. The authors created their own dataset consisting of 10 different hand gestures, such as opening WhatsApp, PowerPoint presentation, Microsoft Word, Microsoft Edge, Google Chrome, Video Player, Xbox, and Paint, etc. They collected a total of 3000 images for training, 2000 images for testing, and 500 images for validation. To train the hand gesture recognizer model, the authors used convolutional neural network (CNN) architecture revised from Squeezenet and VGGNet. They performed image data augmentation using the Keras ImageDataGenerator to augment images in real-time while the model is still training. They used the VGG19 architecture for creating the model, which is a deep convolutional neural network having convolution layer, pooling layer (max pool), fully connected layer, ReLU and softmax in its architecture. They used the Adam optimizer for better accuracy. After the model was trained, they compiled and tested the model to recognize hand gestures and perform the corresponding operations. The captured image of the hand gesture is preprocessed, and a hand detector tries to filter out the hand image from the captured image.

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5. Controlling Web Browser Via Different Hand Gestures And Speech Recognition

This paper presents a system that captures grayscale images from a web camera connected to a laptop as input. Grayscale images are preferred over color images because they require less information for each pixel. The captured image is pre-processed by applying a Gaussian blur to reduce noise, and Otsu's Method is used to automatically select a threshold value based on the shape of the histogram of the image. Contours, which are curves joining all the continuous points along borders having the same color or intensity, are utilized for shape analysis and object detection. Convexity defects, which are any deviation of an object from its convex hull, are detected using contour analysis for finger count extraction. The depth of these defects provides information about the position and orientation of the hand, which is useful for hand segmentation and detection. Additionally, speech recognition is used to convert spoken commands into corresponding text for controlling the web browser. This system has potential applications in hands-free browsing and accessibility for individuals with disabilities.

6. Web Browser Control Through Hand Gesture Using Web RTC API

The system with the help of hardware devices such as a web camera or microphone, this live video feed of user interaction with a webcam is captured with the help of Web RTC API. This structure will be the main interface for user and browser to use the Gesture API. This API will incorporate a custom algorithm that is going to decide the nature of hand motion captured through webcam using Web RTC API. The motion captured will be considered as a live video feed which will be piped to the Canvas tag. An algorithm will analyse every frame as to detect the hand motion and translate it into a custom trigger which will be incorporating a function of user's choice. To implement the algorithm, Canvas tag will divide the individual frame captured from video feed in nine quadrants. Now the algorithm will convert the RGB frame into black and white frame. The white part will represent the skin of hand and rest of the objects will be considered as the black region. Now every frame will be analyzed for the white region and the difference obtained in the nine quadrants will decide the up, down, left and right actions.

III. PROPOSED SYSTEM

Our proposed project leverages two deep learning models, namely the BlazePalm detector model and the Hand landmark model, to accurately and efficiently detect hand gestures in real-time. The BlazePalm detector model identifies the presence of palms in an image, while the Hand landmark model predicts the location of 21 key points on the hand.

To enhance gesture recognition, we develop a custom mathematical model that utilizes the Hand landmark model to identify and recognize complex gestures based on specific landmarks. This model can be easily customized to identify new gestures as well.

To make the system user-friendly and easily accessible, we develop a Chrome extension that allows users to perform various actions on a website based on the detected gestures. This extension is designed to function across all Chromium-based browsers and can be extended to other browsers as well. The proposed system is entirely run locally on the user's computer, which ensures the privacy and security of user data. The low latency on gesture recognition due to local processing, combined with this security feature, makes the system an efficient and secure means for users to interact with websites using hand gestures.

A. Hand Pose Detection

We are using the pre-trained model of the MediaPipe implemented using TensorFlow JS. Which consists of two models that work together. The first model is a palm detector that operates on a full input image and locates palms via an oriented hand-bounding box. The second model is a hand landmark model that operates on the cropped hand bounding box provided by the palm detector and returns high-fidelity 2.5D landmarks. The accurately cropped palm image reduces the need for data augmentation and allows the network to dedicate most of its capacity toward landmark localization accuracy.

The palm detector model uses a single-shot detector optimized for mobile real-time applications, similar to BlazeFace, to detect initial hand locations. The model is trained to detect palms instead of hands, as estimating bounding boxes of rigid objects like palms and fists is significantly simpler than detecting hands with articulated fingers. The model uses an encoder-decoder feature extractor for larger scene-context awareness,

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even for small objects. The focal loss is minimized during training to support a large number of anchors resulting from the high scale variance.

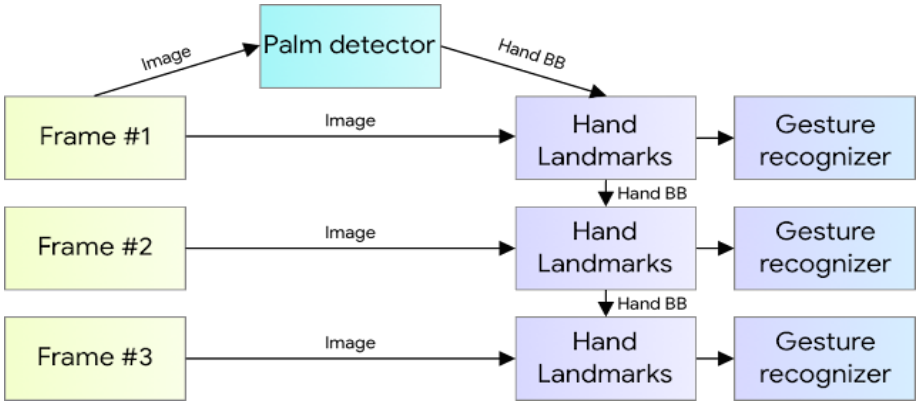
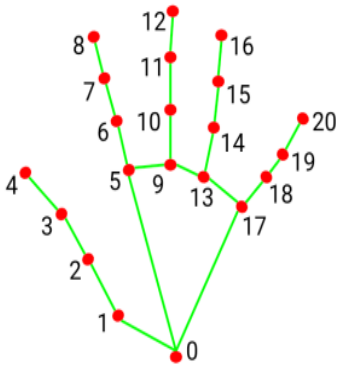


Fig 1. Hand Pose Detection Flow Chart

The hand landmark model performs precise landmark localization of 21 2.5D coordinates inside the detected hand regions via regression. The model learns a consistent internal hand pose representation and is robust even to partially visible hands and self-occlusions. The model has three outputs: 21 hand landmarks consisting of x, y, and relative depth, a hand flag indicating the probability of hand presence in the input image, and a binary classification of handedness (left or right hand).



0. WRIST
1. THUMB_CMC
2. THUMB_MCP
3. THUMB_IP
4. THUMB_TIP
5. INDEX_FINGER_MCP
6. INDEX_FINGER_PIP
7. INDEX_FINGER_DIP
8. INDEX_FINGER_TIP
9. MIDDLE_FINGER_MCP
10. MIDDLE_FINGER_PIP
11. MIDDLE_FINGER_DIP
12. MIDDLE_FINGER_TIP
13. RING_FINGER_MCP
14. RING_FINGER_PIP
15. RING_FINGER_DIP
16. RING_FINGER_TIP
17. PINKY_MCP
18. PINKY_PIP
19. PINKY_DIP
20. PINKY_TIP

Fig 2. Hand Landmark Points

B. Mathematical Model

Finger	Curl	Direction
Thumb	No Curl	Diagonal Up Right
Index	Full Curl	Horizontal Right
Middle	Full Curl	Horizontal Right
Ring	Full Curl	Horizontal Right
Pinky	Full Curl	Horizontal Right



Fig 3. Thumbs-Up Gesture Prediction

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The mathematical model takes the hand landmark points as input and returns all gestures that are made with accuracy. This approach involves defining specific finger positions, angles, and directions for each custom gesture, which are used by the mathematical model to accurately detect and recognize gestures. For example, the "Thumbs Up" gesture can be described as "All four fingers fully curled and pointing to either the left or right. The thumb must not be curled and point upwards." This improved representation allows for a more effective comparison of different hand gestures. The gesture configuration involves defining the specific finger positions, angles, and directions that constitute a particular gesture.

The following finger states are calculated and compared with the gesture configuration.

Table 1: Types of Finger Curls

S. No	Curl
1	No Curl
2	Half Curl
3	Full Curl

TABLE 2: Types of Directions of Fingers

S. No	Direction
1	Vertical Up
2	Vertical Down
3	Horizontal Left
4	Horizontal Right
5	Diagonal Up Left
6	Diagonal Down Right
7	Diagonal Down Left

TABLE 3: Types of Swipe Directions

S. No	Swipe Direction
1	Up
2	Down
3	Left
4	Right

The mathematical model uses a series of conditional statements to compare the landmark points of the hand with the expected positions for each gesture. For instance, to detect a swipe gesture, the model checks consecutive frames to see if the hand has moved in a particular direction. If the hand moves in that direction for a certain distance, the model recognizes it as a swipe gesture. The model measures the angle between two adjacent points on the finger to detect the curl of the fingers and compares it to a threshold value. If the angle is less than the threshold, the finger is considered curled. Similarly, to detect the direction of the fingers, the model measures the overall angle of the hand and compares it to a set of expected values for each direction.

C. Chrome Extension

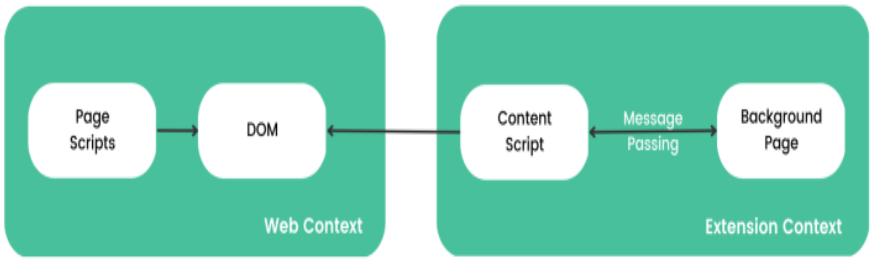


Fig 4. Chrome Extension Block Diagram

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Chrome Extension offers the ability to load and predict models directly within the browser, without the need for any external tools or applications. It also provides a custom pointer on the screen that can perform actions such as click, swipe, zoom in, zoom out, scroll, and even custom interactions. In addition, it allows the user to manipulate the Document Object Model (DOM) to control the website pointer and execute necessary actions. This module can be easily installed using the Extension store in the browser and employs caching mechanisms to optimize the performance of the models. Importantly, this module runs in the background without affecting the performance of the web page, making it an efficient and useful tool for browser-based applications.

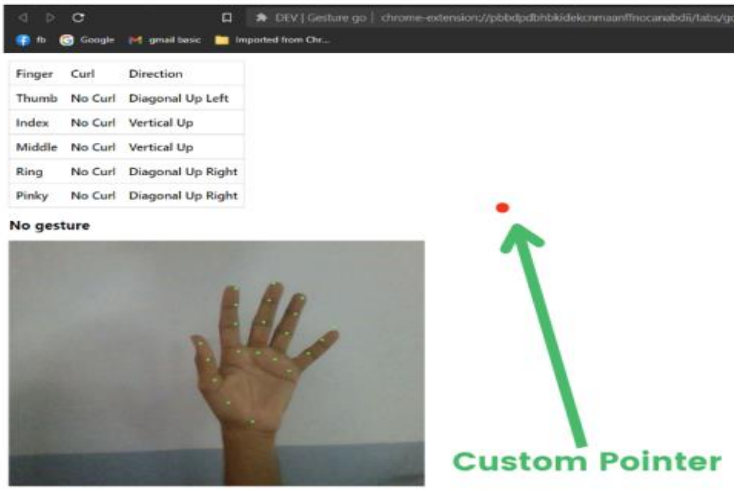


Fig 5. Prediction of Hand And Controlling The Custom Pointer

IV. ADVANTAGE OF PROPOSED SYSTEM

The hand gesture recognition system is designed with several features that enhance its usability and security. Firstly, the system maintains the security and privacy of user data as all processes are executed within the user's computer. Additionally, the application is built as a Chrome Extension, which can be easily installed on all chromium-based browsers, and does not require external dependencies, servers, or networks to function. As video processing occurs within the system, there are no latency issues during gesture recognition. Furthermore, the integration of a mathematical model increases the accuracy of gesture detection. The system is also designed with a modular architecture, making it highly reusable and easily extendable to other browsers. These features collectively make the hand gesture recognition system a secure, efficient, and easily installable solution for enhancing the user experience on the web.

V. CONCLUSION

The described Chrome extension offers users the ability to control website interactions through hand gestures captured by their device's webcam. With all processing done locally, users benefit from increased security and privacy, while experiencing seamless and accurate gesture recognition. Built with a modular design, the extension is highly reusable and can be extended to other applications. Deep learning and mathematical models contribute to increased performance and accuracy of gesture recognition. The extension holds potential for applications in fields such as gaming, virtual reality, and remote control of devices. Gesture control also enhances user engagement and interactivity, resulting in a more immersive browsing experience. Furthermore, gesture control opens new possibilities for web development, allowing for innovative and creative user interfaces and experiences.

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